

WINTERLAND, NL, Canada

WMO: 711810

Lat: 47.1365N

Lon: 55.3265W

Elev: 96

StdP: 14.65

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	12.1	0.7	5.7	11.4	3.9	6.7	13.7	45.5	28.4	41.1	26.0	12.9	300	0.775

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	74.3	65.7	72.1	64.4	69.9	63.5	68.4	71.5	66.7	69.7	65.2	68.1	13.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.2	70.2	65.5	94.5	68.3	63.9	89.3	66.9	32.7	71.5	31.4	69.9	30.2	68.1	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.2	31.8	28.2	DB	3.5	79.3	5.0	2.5	-0.1	81.1	-3.1	82.5	-5.9	83.9	-9.5	85.7	(4)
(5)			WB	1.8	71.6	4.8	1.1	-1.7	72.3	-4.5	73.0	-7.2	73.6	-10.6	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	26.2	25.1	29.1	37.1	45.3	52.8	61.3	62.8	56.1	47.8	39.7	32.3		
	DBStd	14.11	7.00	7.39	6.45	5.08	4.79	5.36	4.26	4.03	5.76	6.09	6.51	7.11		
	HDD50	3650	737	697	648	387	161	30	0	0	12	117	314	547		
	HDD65	8038	1202	1117	1113	837	611	367	128	90	269	533	759	1012		
	CDD50	1125	0	0	0	0	15	113	351	396	196	49	5	0		
	CDD65	39	0	0	0	0	0	1	13	21	4	0	0	0		
	CDH74	72	0	0	0	0	0	1	32	35	4	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	13.0	15.2	16.0	14.8	13.7	12.3	11.4	10.1	10.0	11.5	12.9	13.0	14.8		
(15) Precipitation	PrecAvg	58.8	5.1	4.7	4.4	4.5	4.7	3.7	4.1	3.7	5.4	6.4	6.1	5.8		
	PrecMax	68.9	7.9	7.2	9.1	11.4	7.6	6.5	7.1	5.4	8.8	10.5	9.1	8.0		
	PrecMin	51.3	3.5	2.5	2.5	1.9	1.6	1.4	2.2	2.4	1.8	3.4	3.0	3.8		
	PrecStd	4.8	1.1	1.3	1.7	2.5	1.6	1.5	1.2	0.8	2.0	1.9	1.6	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	45.8	49.1	55.5	66.0	72.9	77.5	77.3	73.7	65.6	58.7	54.0		
		MCWB	50.2	45.2	46.4	48.2	56.4	62.9	68.4	67.4	65.9	62.8	57.0	53.3		
	2%	DB	44.0	41.7	44.0	51.6	61.6	69.1	74.1	74.6	70.1	63.1	55.6	49.8		
		MCWB	43.6	41.1	41.6	47.1	52.8	60.8	65.4	66.3	64.6	60.7	54.3	49.0		
	5%	DB	38.9	37.9	40.6	48.5	57.8	66.0	71.8	72.5	67.6	60.4	53.3	46.1		
		MCWB	37.8	36.8	38.3	44.8	50.5	58.0	63.8	65.1	63.3	57.5	52.2	45.3		
	10%	DB	35.8	35.1	37.9	45.3	54.5	62.7	69.4	70.2	65.4	57.6	50.6	42.8		
		MCWB	34.4	33.7	35.6	41.9	49.0	56.1	62.7	64.2	61.4	54.8	49.2	41.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	45.3	46.8	51.8	59.2	66.2	70.3	70.8	68.8	63.8	57.4	53.8		
		MCDB	50.2	45.5	48.3	54.0	62.9	70.6	75.4	74.2	71.0	64.9	58.3	54.3		
	2%	WB	43.5	40.8	42.1	48.4	55.1	62.5	67.9	68.8	66.3	61.3	54.6	49.1		
		MCDB	43.8	41.0	43.8	50.4	58.2	66.4	71.7	71.7	68.7	62.5	55.5	49.7		
	5%	WB	38.1	36.7	38.6	45.5	52.2	60.0	65.8	67.2	64.3	58.1	52.5	45.4		
		MCDB	38.6	37.4	40.0	47.8	55.9	64.2	69.3	70.0	66.8	59.8	53.3	46.0		
	10%	WB	34.7	33.7	35.8	42.2	49.8	57.5	64.2	65.7	62.2	55.3	49.3	41.8		
		MCDB	35.6	34.5	37.4	44.4	54.1	61.6	67.2	68.6	64.6	57.2	50.3	42.6		
(35) Mean Daily Temperature Range	MDBR		10.5	10.9	11.3	12.9	15.5	15.4	14.6	13.6	14.2	12.2	11.5	10.1		
	5% DB	MCDBR	15.4	14.2	14.4	17.2	21.4	20.1	18.1	16.4	15.1	14.2	14.8	14.3		
		MCWBR	16.2	15.0	12.8	13.1	14.1	12.6	10.7	9.8	10.3	11.8	14.8	14.9		
	5% WB	MCDBR	15.2	14.2	14.2	16.2	18.2	17.4	14.7	13.6	13.7	13.6	14.5	14.6		
		MCWBR	16.1	14.9	13.8	13.9	13.6	12.4	10.0	9.8	10.7	12.8	15.0	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.319	0.332	0.372	0.379	0.402	0.430	0.406	0.371	0.352	0.324	0.313		
	taud		2.341	2.425	2.431	2.343	2.385	2.349	2.320	2.388	2.465	2.466	2.472	2.385		
	Ebn at Noon		239	270	283	280	281	274	264	266	266	254	239	228		
	Edh at Noon		23	27	31	37	37	38	39	35	30	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		321	543	911	1330	1593	1631	1554	1378	1124	699	388	267		
	RadStd		24	55	56	116	120	166	113	85	71	51	48	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page